

# SIDDHANTA MOHANTY

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## ABOUT ME

Results-driven Computer Science senior specializing in AI/ML, backend development, and scalable software systems. Delivered production-ready LLM/RAG pipelines, computer vision algorithms, and backend architectures that cut resolution time by 50% and reduced debugging effort by 40%. Skilled in Python, C++, Flask, LangChain, and PostgreSQL, with a proven ability to turn complex problems into high-impact, maintainable solutions.

## EDUCATION

**Pennsylvania State University - University Park** August 2022 - May 2026  
*Bachelor's, Computer Science*  
• CMPSC 465: Data Structures and Algorithms; CMPSC 464: Introduction to the Theory of Computation; CMPEN 331: Computer Organization And Design; CMPSC 442: Artificial Intelligence; CMPSC 473: Operating Systems Design & Construction

## CERTIFICATIONS

AI For Everyone – Coursera Issued by: DeepLearning.AI Completion Date: [May, 2024] - Gained foundational knowledge of **AI, machine learning, and deep learning terminology** and applications - Learned to identify realistic AI applications, limitations, and ethical considerations in organizational contexts

## SKILLS

- **Languages & Web:** Python, C++
- **Frontend:** HTML/CSS, Tailwind CSS, JavaScript
- **Backend & Frameworks:** Flask, REST APIs, Node.js
- **AI & Machine Learning:** RAG, Agentic RAG, Vector Databases (Pinecone, FAISS), LLMs (GPT-4, OpenAI API), MCP (Model Context Protocol), Tensorflow, NumPy
- **Databases:** MySQL, PostgreSQL
- **Development Tools:** Git, Docker, VS Code, Postman, Streamlit, GitHub, Google Cloud Platform
- **Other:** GDB, Cursor IDE

## PROFESSIONAL EXPERIENCE

**HCL Technologies** Redmond, WA, USA  
*GenAI Intern* May 2025 - August 2025  
• Reduced support resolution time by 50% by shipping an agentic self-healing backend (Flask, LangChain, FAISS) that uses dual-vector RAG over error logs + internal docs, 7-type error categorization, and severity-aware Slack alerts.  
• Automated complex, multi-step workflows across APIs, file systems, and databases via a Model Context Protocol (MCP) system using LangChain agents.  
• Designed a Confluence ingestion pipeline integrated with MCP and LangChain, enabling RAG-based contextual search across organizational documentation to improve developer efficiency.

**Penn State Advanced Vehicle Team** State College, PA, USA  
*Member* February 2025 - Present  
• Worked as part of the Perception Team to build ROS2-based algorithms detecting and tracking traffic lights, signs, road markings, parking spaces, and static/dynamic obstacles with >90% accuracy.  
• Developed multi-sensor fusion and localization systems combining LiDAR, radar, camera, and IMU to ensure accurate navigation under degraded GPS conditions.  
• Contributed to competition-ready frameworks including CAN message integration and perception pipelines aligned with SAE Year 4/5 challenge requirements.

**PennState University** University Park, PA, USA  
*Research Assistant* September 2025 - Present  
• Research Assistant on a project integrating Reinforcement Learning (RL) into a multi-stage large language model (LLM) system for code and content verification.  
• Designed the RL layer that leverages reports from a secondary LLM to generate improved prompts and feedback signals for the primary LLM.  
• Worked within a pipeline where outputs from the main LLM are analyzed by compilers and static analysis tools (Compiler, Klee, CodeQL) before refinement through RL.

**Elevatoz Loyalty** Bengaluru, KA, India  
*Intern* June 2024 - August 2024  
• Applied SQL and Python (pandas, NumPy) to extract, manipulate, and analyze large datasets for business intelligence and customer insights.

## PROJECTS

**Self-Heal System** June 2025 - Present  
• AI self-healing pipeline: logs rich context to PostgreSQL, embeds in FAISS, retrieves similar incidents, and generates GPT-3.5 fixes via LangChain.  
• Severity-aware Slack alerts + 7-type error taxonomy with auto-fixable tagging; deep-link to chatbot for rapid triage.

**Photobot** May 2025 - Present  
• Created an AI-powered photography assistant using RAG + LangChain to provide DSLR/mirrorless settings tailored to context (e.g., astrophotography, sports).  
• Cut configuration decision time by 60% through semantic retrieval from PDF manuals.

**Swipeflix** January 2025 - Present  
• Built a personalized movie/TV recommender using quiz-based profiling + real-time swipe feedback, improving recommendation accuracy over time with TensorFlow.js.  
• Developed a responsive React frontend with smart filters for language, age rating, and genre.

**Prompt Tracer** July 2025 - Present  
• Designed a browser extension to optimize AI prompts across ChatGPT, Claude, Grok, and Gemini with real-time scoring and recommendations.  
• Implemented privacy-first architecture with local processing and cross-platform integration.

**HackPSU(Oct 2024)**  
*CourseScheduler*  
• Developed an NLP-powered academic planner with Node.js + MongoDB backend and React/Tailwind CSS frontend for conflict-free graduation scheduling.

## HONORS/AWARDS

Dean's List Fall 2023, Spring 2024